MIL-L-45958A(AL)
31 May 1983
SUPERSEDING
MIL-L-45958(WC)
13 September 1973
(See 6.4)

MILITARY SPECIFICATION

# LATHE, ENGINE, FLOOR-MOUNTED, 28-INCH SWING

This specification is approved for use by the US Army Armament Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of The Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a rigid base, floor-mounted, metal-cutting, electric motor driven, 28-inch minimum swing engine lathe with four-way power rapid traverse, standard equipment, and accessories as specified herein.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| CC-M-1807 | - | Motor, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller) |
| GGG-C-350 | - | Chuck, Drill; and Arbors and Keys, Drill Chuck |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Materiel Readiness Command, ATTN: DRSAR-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3416

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MILITARY

MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-M-18058 - Machinery, Metal and Wood-Working, Support Equipment and Associated Repair Parts; Preparation for Delivery of
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating, Solvent Containing

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards For Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-889 - Dissimilar Metals

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B5.1 - T-Slots - Their Bolts, Nuts, Tongues and Cutters
B5.8 - Chucks and Chuck Jaws

B5.9 - Spindle Noses for Toolroom Lathes, Engine Lathes, Turret Lathes, and Automatic Lathes
B5.10 - Machine Tapers, Self-Holding and Steep Taper Series
B5.16 - Accuracy of Engine Lathes and Toolroom Lathes
Z210.1(ASTM E380) - Metric Practice

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

201.02 - Tooth Proportions for Coarse-Pitch Involute Spur Gears
390.03 - Gear Classification Manual
600.01 - Metric Practice



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(Application for copies should be addressed to American Gear Manufacturers' Association, One Thomas Circle, Washington, DC 20005.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 79 - Electrical Standard for Metalworking Machine Tool

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one complete lathe for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The lathe shall be a floor-mounted, screw-cutting type, with quick change gearing, of current standard commercial design. The gear range shall provide instant selection of the correct speeds and feeds. The lathe shall meet or exceed the accuracy requirements for engine lathes specified in American National Standard ANSI B5.16. Parts and assemblies shall be new and of the size, strength, and material necessary to sustain the maximum allowable load imposed upon the components and the lathe, with maximum working efficiency and with minimum wear.

3.2.1 Maintainability. Item configuration and adjustment features shall be designed to reduce maintenance operations to the simplest procedures, compatible with a minimum of maintenance skills and the use of standard maintenance equipment. Instructions for disassembly, repair, adjustment, and reassembly shall be concise and clearly defined.

3.2.2 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry. When dissimilar metals are used in contact with each



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other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.2.3 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.3 Construction. The lathe shall be free from any characteristics or defects which could render the machine unsuitable or inefficient for the purpose intended. The lathe and its equipment shall be complete so that, when installed in its intended application and connected to the proper power source, it can be used for any operation for which it was designed. No process shall be used for reclaiming any parts of the lathe. The required swing over the bed shall not be obtained by raising the headstock and tailstock on a basically smaller machine. The lathe may be constructed of parts manufactured to either the US (customary) or International System of Units (SI metric), with all related technical publications, data, and information plates in necessary agreement.

3.3.1 Castings and forgings. Castings shall be free of blowholes, porosity, hard spots, shrinkage defects, cracks and other defects. Forgings shall be free of scale, inclosures, cold shuts, mismatching, and other defects. Physical and chemical properties of castings and forgings shall be adequate to meet the performance requirements of this specification. Patching processes such as welding, peening, plugging, or filling with cold solder or metallic pastes shall not be used on castings or forgings.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts or assemblies.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Safety devices. The lathe shall be fitted with safety devices for all parts that present safety hazards. The devices shall include guards for belts, pulleys, and guideways. All guards shall be designed to provide access to the guarded parts. When special safety devices are required, they shall be specified by the procuring activity (see 6.2).

3.3.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3.6 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the connections shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness.

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3.3.7 Dials, gages, and plates. Dials, gages, and instruction plates shall be permanently marked with plain bold-faced English letters. Dials and gages shall be graduated in the inch (English) and metric system (see 3.3).

3.3.7.1 Data plates. Data plates shall be provided on the head-stock in full view of the operator and shall clearly and legibly indicate the settings for different screw threads, feeds, and speeds.

3.3.8 Gears. Gears and pinions used in the lathe or its components shall be fabricated in either the English (USA - Customary) or Metric (SI) system of measurement. English gears shall conform to AGMA 390.03. Spur gears shall conform to AGMA 201.02. Metric gears shall comply with the International System of Units (SI), International Standards Organization (ISO) Standard 1000, American National Standard ANSI Z210.1 (ASTM E 380), and AGMA 600.01.

3.3.9 Lubrication. All bearings (except the sealed-for-life type), mating gears, revolving shafts, and other working parts shall be provided with standard means for lubrication. Mechanisms requiring special lubrication shall be in accordance with the requirements specified herein. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, where applicable, shall have means for checking levels.

3.3.9.1 Lubrication chart. A plate or plates, secured to each lathe in a conspicuous position shall indicate the principal points to be lubricated as well as the frequency of application. Also, such data as the type and grade of lubricant to be used on the lathe, in the reservoirs, and the motor shall be specified. Complete oiling instructions shall be furnished with each lathe.

3.3.10 Interchangeability. All replaceable parts shall be designed and manufactured to such standards as shall permit replacement without the need for modification of the parts or the lathe.

3.3.11 Surfaces. All surfaces of the lathe shall be clean and free from sand, dirt, fins, sprues, scale, flux, or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled except as may be acquired by original design.

3.4 Sizes and capacities. The sizes and capacities of the lathe shall be as specified in Table I. The requirements specified therein are considered to be minimum and, unless otherwise indicated, are in inches.

TABLE I. Sizes and capacities.

| Requirements: | Sizes and Capacities |
|---|---|
| Swing over bed and carriage wings | 28 |
| Swing over cross slide | 16 |
| Distances between centers, w/tailstock flush with end of bedways | 72 |
| **Headstock:** | |
| Spindle speed number | 27 (or variable) |



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TABLE I. Sizes and capacities. - Continued.

| | |
|---|---|
| Range of spindle speeds: | |
| Low (RPM), not more than | 18 |
| High (RPM), not less than | 1016 |
| Spindle - size of hole through center | 2-1/16 |
| Spindle nose per ANSI B5.9 | 11-D1 or 11-A1 |
| Carriage: | |
| Compound rest travel | 5-1/4 |
| Cross slide travel without taper attachment | 13 |
| Feeds - threads: | |
| Changes, number of | 55 |
| Feed range: | |
| Low (inch per rev), not more than | 0.0025 |
| High (inch per rev), not less than | 0.068 |
| Thread range: | |
| Low, threads per inch, not more than | 2 |
| High, threads per inch, not less than | 60 |
| Lead screw diameter | 1-1/2 |
| Tailstock: | |
| Spindle diameter | 4 |
| Spindle travel | 9 |
| Tailstock set over, each side of center line | 1/8 |

3.5 Components.

3.5.1 Bed. The lathe bed shall be a casting of uniform close grain cast iron or steel. The bed shall be ribbed or cross-braced to provide resistance to twisting as well as to horizontal and vertical deflection. The bed width at the ways shall not be less than 19.5 inches. The bed base width shall not be less than 70 percent of the total bed width measured at the bed ways.

3.5.1.1 Bed ways. Bed ways supporting the carriage shall be flame or induction hardened or shall have hardened and ground steel inserts. The bed bearing surface shall consist of two vee ways and two flat ways, three vee ways and one flat way, or four vee ways. Bearing surfaces shall be uniform with no deviation from a plane surface by more than 0.0005 inch per foot over the entire length. Hardened bed ways shall have a Rockwell C hardness of at least 50. Inserted steel ways shall have a Rockwell hardness of not less than 60 on the C scale.

3.5.2 Headstock. The headstock shall be totally enclosed and contain all gearing for speed ranges as specified in Table I. The headstock shall insure strength and rigidity.

3.5.2.1 Spindle speed change mechanism. Spindle speed changes shall be provided as shown in Table I. The gear train for the spindle drive shall prevent conflicting ratios from engaging. The gear selector shall hold the gears in mesh and shall not move except by positive action of the operator.



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3.5.2.2 Headstock spindle. The inner and outer wearing surfaces of the spindle nose shall have been hardened. The taper per foot of the hole in the spindle shall conform to ANSI B5.10 and shall be finish ground. The spindle nose shall be in accordance with ANSI B5.9. The rotation of the spindle shall be reversible.

3.5.2.3 Spindle controls. The spindle operating controls shall be located on the headstock and on the apron or carriage, making it unnecessary for the operator to leave his working position to start or stop the rotation of the spindle.

3.5.2.4 Headstock lubrication. Headstock lubrication shall be obtained through a pressure system or splash system which shall provide a full flood of oil over all gears and moving parts while the lathe is in operation.

3.5.2.5 Headstock spindle bearings. The spindle shall be mounted on not less than three precision type ball or roller bearings. When properly installed in the headstock, bearings shall withstand the loads imposed upon them when the lathe is operated at its full capacity without affecting the accuracy of the lathe.

3.5.2.6 Quick change gear box. All gearing between the spindle and the lead screw shall be enclosed. The selection of feeds and threads shall be readily obtained by shifting gears inside the gear box by controls located on the front of the enclosed gear box. The gear box shall contain the gears necessary to provide the quantity of feeds and threads specified in Table I.

3.5.3 Tailstock. The tailstock shall comply with Table I and shall have bearing surfaces on both sides of the bed for its entire length. A sensitive screw adjustment shall be provided to set over the tailstock for alignment and taper turning. The tailstock shall be readily removable from the bed to increase carriage travel of length capacity. The base of the tailstock shall be finished for accurate alignment with the headstock as specified in ANSI B5.16. A positive clamp or locking device shall be provided for holding the tailstock at any position on the lathe bed. Loosening and tightening the clamp shall not cause misalignment with the headstock center. A handcrank operated pinion and rack shall facilitate easy movement of the tailstock along the bed.

3.5.3.1 Tailstock spindle. The tailstock spindle shall be accurately ground to fit the housing. It shall be provided with a No. 5 taper in accordance with ANSI B5.10. The tailstock center shall be self holding and provided with a means for ejecting a center using a tailstock hand wheel or knockout rod.

3.5.4 Carriage. The carriage shall be fitted to the bed and provided with adjustable gibs to compensate for wear. There shall be provisions on the carriage for clamping it in place on the bed. A safety clutch or other suitable means shall be provided to prevent damage to the feed mechanism if the carriage is traversed against the headstock or clamped tailstock.

3.5.4.1 Apron. The apron shall provide inboard and outboard bearing supports for all shafts. Gearing in the apron shall provide travel for both the carriage and cross slide. The control for both the carriage and cross slide shall be located on the front of the apron. The carriage feed and threading mechanism shall be constructed so that they cannot be accidentally engaged at the same



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time. A positive interlock shall be furnished to prevent using one when the other is in operation. A neutral position shall be provided when neither the longitudinal feed or threading mechanism is being used.

3.5.4.1.1 Power rapid traverse. Four-way power rapid traverse control(s) for the carriage and the cross slide shall be located on the apron in a position convenient to the operator. Gear protection of the lathe shall be provided by interlocks to prevent the longitudinal and cross feed and traverse from engaging at the same time.

3.5.4.2 Carriage reversing mechanism. The reversing mechanism shall afford movement of the carriage in either direction without changing the direction of rotation of the spindle. The reversing control shall be located on the apron in a position convenient to the operator.

3.5.4.3 Cross slide. The cross slide shall be fitted to the carriage and shall have a means for compensation of wear.

3.5.4.4 Lead screw. The lead screw shall provide the neccesary stability and retention of accuracy within the limitations specified herein. The lead screw shall have a minimum diameter of 1.5 inch. The lead screw error shall not exceed that specified in ANSI B5.16.

3.5.4.5 Feed screws. The cross feed and compound feed screws shall provide the necessary stability and retention of accuracy. The cross feed and compound feed dials shall be graduated to indicate 0.001 inch and 0.02mm stock removal from the work diameter. Adequate means of lubrication for feed screws shall be provided. The cross feed screw shall be arranged to permit the use of the taper attachment.

3.5.4.6 Thread cutting lead screw. Screw thread cutting leads shall cover all standard thread pitches in accordance with FED-STD-H28. Thread range shall be as specified in Table I.

3.5.4.7 Thread chasing dial indicator. A thread chasing dial indicator shall be provided and conveniently located on the carriage or apron and shall be graduated to indicate correct engagement of the lead screw.

3.5.4.8 Automatic carriage stops. An adjustable automatic stop shall be provided to stop the travel of the carriage at any predetermined point between the carriage and the headstock.

3.5.4.9 Compound rest. The compound rest shall consist of a base and top. The compound rest shall be attached to the cross slide. Means shall be provided for clamping the compound rest without marking or deforming the cross slide mechanism. The compound rest base shall be graduated from zero, as aligned at right angles with the bed ways, in increments of one degree up to and including 90 degrees in both directions from zero. The compound shall swivel 360 degrees without removal of any parts of the cross slide. The compound shall be provided with means for mounting a tool post.

3.5.5 Electrical equipment. Electric motors and control equipment shall be mounted to the lathe and shall operate on 230/460 volts, three phase, 60 hertz (Hz), (initially wired for 230), shall be fungus controlled, and shall be in accordance with NFPA No. 79. The main drive motor shall not be less than 20



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horsepower, induction squirrel-cage type, continuous-duty, drip-proof, ball or roller bearings, 50° centigrade rise over ambient and shall conform to CC-M-1807. Other motors shall be induction squirrel-cage type, continuous duty, conforming to CC-M-1807.

3.6 Standard equipment. The following equipment or accessories shall be furnished with each lathe as standard equipment. The requirements specified are considered minimum.

3.6.1 Lathe centers. Two lathe centers with No. 5 taper shall be furnished. Centers shall conform to ANSI B5.10. The centers shall be hardened and ground to within 0.0002 inch total indicator reading for concentricity and alignment.

3.6.2 Center sleeve. The headstock center sleeve shall be provided with a means for adapting from the taper in the spindle nose to the taper of the centers supplied. Concentricity shall be within 0.0002 inch total indicator reading. The dimensions and tolerances shall be in accordance with ANSI B5.10.

3.6.3 Small face plate. The small face plate shall fit the spindle nose of the lathe. The face plate shall have two or more driving slots. The face plate size shall have a 12-inch diameter. Accuracy and run out shall be in accordance with ANSI B5.16.

3.6.4 Large face plate. The large face plate shall fit the spindle nose of the lathe. The face plate shall have four or more T-slots or through slots in accordance with ANSI B5.1. The face plate size shall have a 24-inch diameter. Accuracy and run out shall be in accordance with ANSI B5.16.

3.6.5 Steady rest. The steady rest shall fit the ways of the lathe. The steady rest shall be of the hinged type with three adjustable tip jaws. All working parts shall be finished and shall contain adjusting and locking screws for each of the three jaws. Steady rest capacity shall be from one to six inches. Means shall be provided for clamping the steady rest to the bed ways.

3.6.6 Tool post. The tool post shall be complete with rocker support and ring and shall be slotted to receive tool shanks, or tool holder shanks, 0.875 by 1.75 inches.

3.6.7 Micrometer stop. The micrometer carriage stop shall be graduated in thousandths of an inch. It shall have a working range of not less than one inch.

3.6.8 Follow rest. The follow rest shall be fitted to the carriage with means for quick installation and removal. It shall have two or more adjustable tip jaws. The rest shall have adjusting screws and lock screws for setting and locking the jaws for one to six-inch diameters.

3.6.9 Taper attachment. The taper attachment shall be of the telescope screw-type, or universal-type, with a 3.5-inch taper turning capacity per foot and an 18-inch length capacity at one setting. It shall be mounted on the rear of the bed or on the carriage. The attachment shall be graduated in both inches per foot and degrees. Locking screws shall be provided for maintaining adjustment. The attachment shall not interfere with regular operation of the lathe when the attachment is not in use.



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3.6.10 Independent four-jaw chuck. The chuck body shall be semi-steel, heavy duty, with finished back and fitted to the spindle of the lathe. It shall be of the individual screw adjusting type in accordance with ANSI B5.8 for Class I, 18-inch chucks. Chuck jaws shall be reversible, alloy steel with three ground steps and bite, hardened to resist wear, chipping, or breakage, and fitted to the chuck body. Test procedures and tolerances shall conform to ANSI B5.16.

3.6.11 Self-centering three-jaw chuck. The chuck body shall be semi-steel, medium duty, in accordance with ANSI B5.8 for Class I, 15-inch chuck. It shall have a finished back and shall be fitted to the spindle nose of the lathe. Chuck jaws shall be reversible, alloy steel, three ground steps and bite, hardened to resist wear, chipping, or breakage, and fitted to the chuck body. Test procedures and tolerances shall conform to ANSI B5.16.

3.6.12 Coolant system. The coolant system shall include a pump or reservoir, a power-driven pump, and piping. The coolant reservoir shall incorporate a strainer to remove cutting chips from the coolant. The reservoir shall be equipped with a drain plug. The coolant system shall include oil resistant hoses, tubing, or piping to direct the coolant. The coolant delivery nozzle shall be fitted with a valve to permit the operator to control the amount of coolant flow.

3.6.13 Chip and oil pan. A chip and oil pan of adequate strength shall be furnished as part of the lathe. The chip pan shall be free from all sharp edges which could injure the operator. The pan shall be leakproof and shall be arranged for drainage and installation of a coolant system. A drain hole and plug shall be provided.

3.6.14 Wrenches. One set of special wrenches necessary for normal operation shall be provided with each lathe.

3.7 Additional equipment. When specified (see 6.2), the following equipment shall be furnished in quantities of one each with each lathe. Sizes listed are to be considered minimum.

(a) Dog, set, lathe, bent tail, single screw, 2, 3, 4, and 6 inch capacities.
(b) Holder, lathe tool, knurling, self centering head, diamond pattern set of medium knurling tools, shank to be 0.875 inch by 1.75 inches.
(c) Holder, set, lathe tool, cutoff with 0.187 by 1.125 inch blade, consisting of three holders; one right hand offset, one left hand offset, and one straight, shanks to be 0.875 inch by 1.75 inches.
(d) Holder, set, lathe turning tool, consisting of three holders; one right hand offset, one left hand offset, and one straight, shanks to be 0.875 inch by 1.75 inches.
(e) Holder, threading tool for formed cutter, complete with hand speed, sharp, 60 degree angle V thread cutter and wrench, shanks to be 0.875 inch by 1.75 inches.
(f) Holder, boring tool, 0.875 inch by 1.75 inches, 1.312 inch bar diameter with 30, 45, and 90 degree end caps, and three cutters.
(g) Drill chuck, heavy duty, with key 0.375 to 1 inch capacity, in accordance with GGG-C-350, type 1, class 1, unit A. The chuck arbor shall fit No. 5 spindle.



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3.8 Fungus control. When specified (see 6.2), electrical components and circuit elements shall be treated for fungus resistance in accordance with the type II requirements of MIL-T-152, except that the statement of treatment is not required. The windings of the electric motors shall be given two impregnating coats of varnish conforming to the requirements of MIL-I-24092, type M, class 155 through 220 (manufacturer's option).

3.9 Painting. Unless otherwise specified, painting and finishing of the lathe shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall be clean and free of all foreign matter detrimental to painting and shall have one coat of primer and one coat of enamel, or equivalent, applied.

3.10 Product Identification. The lathe shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.11 Workmanship. Standards of workmanship shall assure that the lathe shall have the strength, stability, and efficient operating characteristics found in standard commercial items and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (c) as follows. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).



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4.4.1 Sampling. When required (see 6.2), sampling for quality conformance inspection shall be as specified in MIL-STD-105 at the Inspection Level and Acceptable Quality Level (AQL) specified in the contract. For packaging inspection (see 4.7), special inspection level S-4 and an AQL of 4.0 percent defective shall be used.

4.5 Product examination. Each lathe shall be examined to determine compliance with the requirements of this specification and the contract. As a minimum, the following characteristics shall be examined:

| CHARACTERISTICS | REQUIREMENT PARAGRAPH |
|---|---|
| Design not as specified | 3.2 |
| Material not as specified | 3.2.2 |
| Construction not as specified | 3.3 thru 3.3.11 |
| Sizes and capacities not as specified | 3.4 and Table I |
| Components not as specified* | 3.5 thru 3.5.5 |
| Standard equipment missing or not as specified | 3.6 thru 3.6.14 |
| Additional equipment missing or not as specified | 3.7 |
| Fungus control not as specified** | 3.8 |
| Painting not as specified | 3.9 |
| Marking not as specified | 3.10 |
| Workmanship defects | 3.11 |

*Certification of compliance required for electrical equipment
**Certification of compliance required for fungus control

4.6 Tests.

4.6.1 Accuracy test. Each lathe shall be tested for compliance with ANSI B5.16, tests 1 through 20.

4.6.2 Operational test. Each lathe shall be operated at no load and, during this time shall be inspected to insure functional operation of all motors, controls, operating mechanisms, and accessories.

4.6.3 Turning test. The lathe shall be tested for compliance with ANSI B5.16, test 21 through test 25.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging of the lathes to determine compliance with Section 5 requirements.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be Level A in accordance with MIL-M-18058 or Industrial in accordance with ASTM D3951-82, as specified (see 6.2).

6. NOTES

6.1 Intended use. The lathe covered by this specification is intended for heavy-duty use in machine shops for fine and rough metal turning and boring and for cutting fine and coarse, right and left hand, screw threads.



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6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Special safety devices when required (see 3.3.4).
(d) Additional equipment when required (see 3.7).
(e) Fungus control when required (see 3.8).
(f) Painting if different (see 3.9).
(g) Identification marking if different (see 3.10).
(h) First article inspection if different (see 4.3).
(i) Quality conformance inspection if different (see 4.4).
(j) Sampling when required (see 4.4.1).
(k) Selection of applicable level of preservation, packing, and marking (see 5.1).

6.4 Supersession data. This specification includes the requirements of Rock Island Arsenal Purchase Description T&EPD-65 dated 8 April 1969.

6.5 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - AL

Review activities:

Army - SM

Preparing activity:

Army - AL

Project No. 3416-A169



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